

FitPulse – AI-Powered Fitness and Nutrition Recommendation System for University Students (P2)

Team Members:

- Nirmala Varun
- Sylvano Dsouza
- Madhukar Goli
- Phuoc Long Hoang

1. Challenges and Mitigation

Challenge 1: Data integration from multiple sources (Kaggle datasets, Streamlit form inputs)

Mitigation: Establishing a unified schema and using pandas-based preprocessing scripts.

Challenge 2: Campus Food Data Inaccessibility: The Pacific website lacked detailed information on food options.

Mitigation: Manually collected the data from campus sources.

Challenge 3: Backend-frontend integration (Streamlit to Github workflow)

Mitigation: Hosting shared Github repo; team using branches to avoid conflicts.

Challenge 4: Task load imbalance and coordination

Mitigation: Weekly syncs, Miro task board, and Google Sheets tracker.

2. New Scripts Created / Modules to Be Developed

1. **Diet Preference Tracker:** Weekly assignment of meals based on dietary inputs (vegetarian, high-protein, etc.)
2. **Data Cleaning Module:** Convert raw food data to structured format for KNN ingestion
3. **Streamlit UI Enhancement:** Improve visual layout, input validation, collapsible sections
4. **Progress Tracker & Notification Timer (Future Scope):** Daily activity log + notification system

3. Member Task Assignment and Plan

Name	Assigned Task	Plan & Tools	Timeline
Nirmala	Project Coordination + Streamlit Form + Feedback Tracker	Streamlit UI build, CSV pipeline, and progress module. Shared via GitLab.	Apr 19–22
Sylvano	Clustering & Persona Segmentation	KMeans modeling using Altair + Python; finalize cluster labels	Apr 18–21
Madhukar	Fitness Recommendation (KNN)	Scikit-learn implementation, test accuracy with A/B personas	Apr 20–22
Phuoc	Nutrition Prediction (Regression)	Linear regression with USDA-comparable values	Apr 19–21

4. Resources Used

1. **Data:** Kaggle (Workout Planner, Body Fat, Nutrition Facts), USDA Guidelines
2. **Tools:** Python, Streamlit, JupyterLab, R (ggplot2), GitLab, Altair Studio
3. **Collaboration:** Google Sheets, Teams, Miro, Canvas

5. Project Plan Expansion

Near-Term Goals:

- Functional Streamlit Form (Input Interface)
- Backend KNN Recommender for Meals/Fitness
- Meal/Workout Plan Display
- Clean UI Design with Visual Enhancements

Future Scope:

- Campus Expansion: Add other university data
- Corporate Use Case: FitPulse for workspaces

- Progress Tracker: Logging goals, updates weekly
- Alerts: Reminders for workouts or meals based on preference

6. Timeline and Progress Overview

Member	Progress	Notes
Nirmala	Regression tuned, final validation running	Ready to connect to frontend
Sylvano	Clusters done, visual testing ongoing	Segment tags pending polish
Madhukar	KNN implemented & evaluated	Evaluating edge case predictions
Phuoc	Streamlit form live, feedback logger functional	Needs export-to-PDF link

Daily Management Tool

[Miro board link](#)

Project Management Sheet

[Project Management Sheet - Group 4](#)